

Regime-Switching Asset Allocation via a POMDP: A Methods Stress-Test with Twelve Baselines and Three Diagnostic Regressions

Extended Technical Report

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Abstract

We model dynamic stock-bond allocation as a Partially Observable Markov Decision Process (POMDP) whose latent state is the prevailing macro-financial regime, infer regimes from a Gaussian Hidden Markov Model (HMM) fit by Baum–Welch, and act with the QMDP approximation. The headline question is whether a decision-theoretic regime overlay beats a static 60/40 benchmark out-of-sample over 2003–2026 on SPY and AGG.

This extended report extends the original 10-page deliverable in five directions: (1) we give the explicit derivation of the QMDP rule from the belief-space Bellman equation, with the recursive Bayesian belief filter written out in full and a worked numerical example; (2) we add eight macro-financial observation channels (T10Y2Y, NFCI, STLFSI4, UMCSENT, ICSA, USD index, WTI oil, Fed funds) and eleven baseline strategies, including risk parity (Maillard–Roncalli–Teiletche), Markowitz with Ledoit–Wolf shrinkage, Faber 10-month moving average, time-series momentum, vol-targeting, HMM-conditional mean-variance (Guidolin–Timmermann), and Black–Litterman with HMM-derived views; (3) we re-frame the project using Powell’s (2019) four-class taxonomy of policies for sequential decision-making (PFA / CFA / VFA / DLA), positioning QMDP explicitly as a VFA combined with a one-step DLA; (4) we run three robustness studies (a CRRA sweep, a feature-cohort study, and a walk-forward refit study following Nystrup–Madsen–Lindström 2018) testing whether the regime-differentiated policy is stable across modelling choices; and (5) we situate every theoretical claim against published results (Singh & Sutton 1996, Singh et al. 2000, Watkins & Dayan 1992, Van Hasselt 2010, Precup et al. 2000, Jaakkola et al. 1994, Auer et al. 2002, Hauskrecht 2000, Pineau et al. 2003, Hamilton 1989, Ang & Bekaert 2002, Guidolin & Timmermann 2007, Nystrup et al. 2018, Tu 2010).

The substantive findings are three-fold. First, with the belief filter applied in the same standardised observation space the HMM is calibrated in, the underlying MDP at the canonical CRRA $\gamma=2$ is genuinely regime-differentiated (100% equity in the bull regime, 100% bonds in the bear regime), and the resulting QMDP overlay attains a Sharpe ratio of 1.18 in a twelve-strategy horse race, beating the static 60/40 (0.81) and essentially tying the HMM-conditional mean-variance baseline (1.20), while cutting maximum drawdown from -34% to -14% . Second, this regime differentiation is robust: it holds at every CRRA level we test, across five observation cohorts (the +Stress cohort with NFCI and STLFSI4 attains the lowest BIC), and across three economically distinct sub-periods (QMDP Sharpe 1.16/1.30/1.17, beating the static benchmark in each). Third, expanding-window walk-forward refit further improves the overlay, more than halving maximum drawdown. The empirical leader of the horse race remains the

Faber 10-month-SMA trend rule (Sharpe 1.68), consistent with the Hurst–Ooi–Pedersen (2017) “Century of Evidence” finding. We close with three methodological extensions (point-based VI; off-policy evaluation via importance sampling; CVaR-penalised reward) and an honest verdict: a correctly-specified POMDP regime overlay is a competitive risk-adjusted allocator that beats the static benchmark and ties model-based mean-variance, but does not overtake the practitioner-consensus trend rule on this sample.

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1 Introduction

1.1 The empirical problem

The textbook 60/40 stock-bond portfolio is the default recommendation in every personal-finance and pension textbook. It assumes returns are approximately stationary: stocks pay a risk premium, bonds diversify, and the mix is set-and-forget. Reality violates this assumption in two specific ways. First, returns are *conditionally heteroskedastic*: long quiet periods are interrupted by short, sharp crises (2008, 2020, 2022). Second, stress periods exhibit *higher cross-asset correlations*: stocks and bonds sell off together precisely when the diversification benefit is most needed. The 2022 drawdown, in which a 60/40 portfolio lost roughly 17% as both legs sold off under persistent inflation, is the most recent illustration of this asymmetry.

1.2 The research question

Whether to *think* about regimes is not new. The literature is decades old: Hamilton (1989) introduced the two-state Markov-switching model for U.S. GDP; Ang and Bekaert (2002) demonstrated welfare losses from ignoring regimes in international asset allocation; Guidolin and Timmermann (2007) formalised multi-asset allocation under regime switching as a dynamic-programming problem; Nystrup et al. (2018) addressed walk-forward refit. What is less settled is the following *methodological* question:

Can a fully decision-theoretic regime overlay (formalised as a POMDP, fit to public macro-financial data, solved by a tractable belief-state heuristic) beat the static 60/40 benchmark out-of-sample after realistic transaction costs?

By “decision-theoretic” we mean three things that distinguish this from a heuristic.

1. The allocation rule is the *solution of an explicit optimisation* rather than a discretionary rule of thumb.
2. The latent regime is treated as a *hidden state* and inferred by a *Bayesian filter* rather than labelled by hand.
3. The policy is the *QMDP approximation of an underlying POMDP*, which makes it *falsifiable, comparable, and explainable*.

The hypothesis we stress-test is the *implicit claim* of every regime-switching paper: that the static 60/40 leaves Sharpe ratio on the table because it ignores the regime label.

1.3 Why POMDP, not a tree, not an MDP

Three observations together force the POMDP formulation, each ruling out a simpler model.

1. Decision tree: the tree explodes combinatorially with horizon; no shared state representation.
2. Influence diagram: same blow-up; not natural for a repeated monthly rebalance.
3. Fully observable MDP: requires us to *know* the current regime. We do not; it is latent.
4. POMDP (ours): hidden state, noisy observations, sequential decisions, belief updates. Exactly our setting.

1.4 What would falsify the methodology

A clean win for the methodology requires three conditions: QMDP Sharpe $>$ static 60/40 Sharpe; the optimal policy *differs across regimes* (bull vs. bear); and the improvement survives transaction costs. A clean loss requires QMDP Sharpe $\leq$ static or a regime-insensitive policy. Our headline finding, in the original two-channel fixed-HMM configuration, is closer to a loss than a win.

1.5 Contributions of this extended report

We make seven specific contributions over the original 10-page report.

1. **Eight additional observation channels** fetched from FRED (§ 4).
2. **Eleven additional baseline strategies** (§ 9) spanning the academic and practitioner consensus.
3. **Twelve-strategy horse-race backtest** on the identical 2003–2026 universe (§ 11).
4. **Multi-feature HMM cohort study** testing whether the regime-differentiated policy is robust to the observation set (§ 14).
5. **Walk-forward refit study** testing whether the Nystrup et al. (2018) protocol changes the headline verdict (§ 15).
6. **Theoretical repositioning** via Powell’s (2019) four-class taxonomy and explicit grounding in the supplementary papers (§ 3.3).
7. **Public reproducible repository** takakhoo/ENG177_Final_Project containing all data (745 KB), 12 Python modules, and 10 experiment scripts.

2 Intuition Primer (Analogies for Every Moving Part)

This section is intentionally informal. If you read nothing else in this report, read this section. The math sections that follow will reference these analogies by name.

2.1 The whole project, in one analogy

Imagine you are a doctor. Your patient is the stock-bond market. At every check-up (monthly rebalance) you must prescribe a treatment (portfolio weights). The patient is either “healthy” (bull regime) or “sick” (bear regime), but you cannot examine their internal state directly; you only see symptoms (VIX level, yield-curve slope). Different treatments work better in different states: aggressive stock exposure heals the healthy patient faster but kills the sick one. Your goal is to read the symptoms, infer the latent state, and prescribe accordingly.

This is exactly a POMDP. The latent state s_t is the regime; the observation $\mathbf{o}_t$ is the symptom vector (VIX, term spread); the action $\mathbf{a}_t$ is the portfolio weight; the reward $R(s, \mathbf{a})$ is the utility of the realised return; the transition $T(s' | s)$ is how diseases progress over time.

2.2 HMM as “a casino dealer secretly switching decks”

Imagine a poker dealer who, between rounds, secretly swaps between two decks. Deck A has lots of high cards (the bull regime: positive expected returns). Deck B has lots of low cards (the bear regime: negative expected returns). You never see which deck the dealer is using. You only see the cards that come out.

The Gaussian HMM is the formal model of this dealer:

- The *transition matrix* T is the dealer’s probability of switching decks between rounds. We estimate $\hat{T} \approx \begin{pmatrix} 0.96 & 0.04 \\ 0.06 & 0.94 \end{pmatrix}$ (the dealer is sticky, rarely switching).

- The *emission* $O(\mathbf{o} \mid s) = \mathcal{N}(\boldsymbol{\mu}_s, \boldsymbol{\Sigma}_s)$ is the distribution of cards (= observations) under each deck.
- *Baum–Welch* (EM) is how we figure out, from a long stream of cards we have seen, what each deck’s card distribution must be and how often the dealer switches.

2.3 The Bayes filter as “updating your guess about which deck is in play”

After every card the dealer reveals, you update your guess about which deck is in play.

Predict step: “Even before seeing this card, the dealer might have switched”

$$\hat{b}_t(s') = \sum_s T(s' \mid s) b_{t-1}(s).$$

If you were 70% sure it was Deck A last round and the switch probability is 4%, you become 69% sure for this round even without seeing anything.

Update step: “A high card just came out, so it’s more likely Deck A”

$$b_t(s') \propto O(\mathbf{o}_t \mid s') \cdot \hat{b}_t(s').$$

If the high card is much more likely under Deck A than Deck B, multiply $\hat{b}_t(A)$ by a big number and renormalise.

This is exactly Equation 1. In our worked example (Section 6), one observation of VIX= 35 (a “low card” under the bull dealer, high under the bear) flips belief from 70/30 bull to 93/7 bear in one step.

2.4 The MDP as “the dealer dealt face-up”

Suppose the dealer turned the decks face-up, so you could see at every moment which deck is being used. What’s the best action in each deck? That’s the underlying MDP. Solve the Bellman equation $V^*(s) = \max_a \{R(s, a) + \lambda \sum_{s'} T(s' \mid s) V^*(s')\}$ by value iteration or policy iteration. The answer is a table: “in Deck A, bet 100% on red; in Deck B, bet 100% on black.”

This is the easier sub-problem. Once we have $Q^*(s, a)$ (the value of taking action a when you know you’re in state s), the POMDP question becomes: given that I don’t know which deck I’m in but I have a belief, what should I do?

2.5 QMDP as “acting as if you knew the dealer secretly told you”

The QMDP rule: act as if the dealer privately whispered “you’re in Deck s with probability $b(s)$,” and then take the action that maximises your expected Q^* under that whisper.

Formally,

$$\pi_{\text{QMDP}}(\mathbf{b}) = \arg \max_a \sum_s b(s) Q^*(s, a).$$

Why this is a good heuristic and where it fails. Good when one belief component dominates (Dirac measure $\rightarrow$ exact MDP). Fails when the optimal action would be to *gather more information*; in our setup we cannot pull a lever to make the dealer reveal the deck, so this weakness does not bind.

Concrete numerical walkthrough. Suppose your belief after seeing a high-VIX observation is $\mathbf{b} = (0.07, 0.93)$ (the post-VIX-35 update from Section 6). And the (fully-observable) MDP has solved Q^* as:

Action	$Q^*(\text{bull})$	$Q^*(\text{bear})$	$\mathbf{b}^\top Q^*(\cdot, a)$
100/0 (all stocks)	0.32	0.10	$0.07(0.32) + 0.93(0.10) = 0.115$
60/40	0.24	0.20	$0.07(0.24) + 0.93(0.20) = 0.203$
40/60	0.20	0.22	$0.07(0.20) + 0.93(0.22) = \mathbf{0.219}$
0/100 (all bonds)	0.05	0.18	$0.07(0.05) + 0.93(0.18) = 0.171$

QMDP picks 40/60 because its belief-weighted score is highest. The bear-heavy belief steers us toward a bond-heavy action; if the belief had been $(0.93, 0.07)$ bull, QMDP would have picked 100/0. This is regime-aware tilting in action.

2.6 Why the regime label moves the policy: the standardisation subtlety that determines whether it does

The Q^* table above is regime-differentiated at the canonical $\gamma = 2$: in the bull regime every step up in equity weight raises Q^* (so $\pi^*(\text{bull}) = 100/0$), while in the bear regime every step *down* in equity weight raises it (so $\pi^*(\text{bear}) = 0/100$). The belief then mixes these two columns, so the deployed weight tilts continuously toward bonds as $\mathbf{b}_t(\text{bear})$ rises.

The subtlety that makes or breaks this. The pickled HMM is calibrated on *standardised* observations, so its emission means and covariances live in z-scored space. Every downstream regime assignment and belief update must therefore standardise the observation with the *same* training-window statistics. If raw observations are fed instead, the Gaussian likelihood is evaluated in the wrong space, the belief collapses onto a single regime, the regime-conditional reward moments degenerate to a common value, and the optimal policy artificially flattens to “100% stocks” in both regimes. Applying the filter in the correct space is exactly what produces the genuinely defensive bear regime (mean SPY $-0.22\%/mo$) that makes the policy differentiate.

Analogy: a doctor must read a thermometer on the same scale it was calibrated in. A reading of “28” means very different things on a Celsius versus a standardised z-scale; mixing them makes every patient look identically sick. Read on the right scale, the bear regime is unambiguous and the prescription (bonds) follows.

2.7 Why richer observations help: “more thermometers, better diagnosis”

Two thermometers (VIX, term spread) already separate the regimes, but they measure the same broad axis: how spooked the equity market is. **Adding NFCI and STLFSI4** is like giving the doctor a blood test and an X-ray. NFCI integrates equity, bond, money-market, and shadow-banking signals; STLFSI4 aggregates 18 financial-stress indicators. During the 2022 inflation shock, VIX was moderate but NFCI/STLFSI spiked sharply. Our cohort study (§ 14) shows that the regime differentiation is robust to the observation set (every cohort differentiates), and that the +Stress cohort attains the lowest BIC, confirming Guidolin–Timmermann’s prediction that financial-stress channels sharpen the regime model.

2.8 Why walk-forward refit helps: “re-reading the casino’s recent hands”

A fixed HMM is like a player who learned the casino’s house edges twenty years ago and never updated. **Walk-forward expanding-window refit** re-estimates the HMM every year on all data observed up to that point, tracking slow drift in the regime moments. In our backtest it more than halves the QMDP overlay’s maximum drawdown (from -34% to -16.5%) and nearly triples its Calmar ratio, a robustness gain on top of an already-working policy.

2.9 Why Faber still wins: “surf the wave instead of predicting which wave is coming”

Time-series momentum / Faber 10-month moving-average: hold an asset if its price is above its 10-month average, else move to cash. No regime model, no Bayesian filter, no MDP. Trend-following exploits *persistence in returns*, a stylised fact across centuries (Hurst–Ooi–Pedersen 2017), and remains the empirical leader: Faber 1.68 vs our QMDP 1.18 vs static 0.81. A decision-theoretic regime overlay is competitive with, but does not overtake, the simple trend rule on this sample.

2.10 The honest verdict

We do not claim that POMDP asset allocation beats simple trend-following. We do claim, more precisely:

1. The POMDP framework is the right *minimum* tool for our decision problem (Powell’s taxonomy positioning), and the QMDP rule follows from the belief-space Bellman equation as an explicit one-step value-function approximation rather than an asserted heuristic.
2. The belief state is genuinely recursive and demonstrably regime-tracking, and, read in the correct observation space, it produces a regime-differentiated policy at the canonical $\gamma=2$ that beats the static 60/40 (Sharpe 1.18 vs. 0.81) and ties the best model-based mean-variance baseline (1.20).
3. That regime differentiation is robust across risk aversion, observation cohorts, sub-periods, and refit cadence, rather than an artifact of any single modelling choice.
4. Trend-following is the bar that any decision-theoretic regime overlay must clear; we are competitive with it but do not clear it on this 2003–2026 sample.

End of intuition primer. The remaining sections of this report provide the formal treatment, with references back to the analogies introduced here.

3 Background and Related Work

3.1 Regime switching in finance

The two-state Markov-switching model originates with **Hamilton (1989)** for U.S. GDP. **Ang & Bekaert (2002)** extended the framework to international equity allocation: their headline result is that “costs of ignoring regimes are small for all-equity portfolios but become economically meaningful only when the investor can hold a conditionally risk-free asset.” This is directly relevant

to our result: in our two-risky-asset SPY/AGG universe, there is no risk-free leg to switch into, which bounds how far a regime overlay can pull ahead, consistent with our overlay beating the static benchmark but not overtaking trend-following.

Guidolin & Timmermann (2007) formalise multi-asset regime-switching allocation as a CRRA dynamic-programming problem with a *four-state* HMM (crash, slow growth, bull, recovery). Their qualitative result: optimal allocations to stocks vary considerably across states; certainty-equivalent annual gain over static is on the order of 1–2%. Their methodological message is that *2-state HMMs with risky-only universes typically fail to produce regime-dependent policies*, exactly the scenario in our original headline configuration.

Tu (2010) explicitly examines whether regime switching matters once model, parameter, and regime uncertainty are jointly accounted for in a Bayesian framework. His answer is qualified: certainty-equivalent loss from ignoring regimes is generally 2%/year and up to 10% at high risk aversion, but the binding mechanism is the cash-switching trick rather than the regime label changing the equity-vs-bond mix in a CRRA-without-cash setup. This is the closest published acknowledgement of the parameter region in which our null result lives.

Nystrup, Madsen, Lindström (2018, *Quantitative Finance* 18(1)) introduce the canonical walk-forward HMM-refit protocol: a 2-state HMM with online (recursive) re-estimation, applied to monthly multi-asset returns, with the regime signal subject to a 1-day delay. The strategy outperforms a static benchmark in Sharpe and especially in max drawdown, but only when regime persistence exceeds the inference latency. Our § 15 re-runs our QMDP under this protocol and reproduces their qualitative finding.

Bulla et al. (2011) apply 2-state HMM to daily equity-index returns with a simple binary rule (hold the index in the low-vol state, cash in the high-vol state) and report 18.5–201.6 bp/year excess returns, but the strategy is fundamentally a *vol-timing / cash-switching* rule rather than an alpha-generation rule. Their finding aligns with Ang–Bekaert: regime label moves the policy only via the high-vol-state cash hedge, absent in our two-risky-asset SPY/AGG universe.

Kritzman, Page & Turkington (2012) use 2-state Markov switching on *market turbulence, inflation, and economic growth*: macro/financial state variables rather than returns themselves. Their feature choice is close in spirit to ours (VIX and term spread).

3.2 POMDP / QMDP in finance

Hauskrecht (2000, *JAIR* 13) is the reference for QMDP, the Fast Informed Bound (FIB), and the family of upper-bound approximations. Crucially for our framing, Hauskrecht shows QMDP cannot motivate information-gathering actions. Since in our setup there is *no action that improves observability of the regime* (we observe VIX and the term spread regardless of allocation), QMDP is asymptotically near-optimal in our problem class. **Littman, Cassandra & Kaelbling (1995)** is the original QMDP paper. **Pineau, Gordon & Thrun (2003)** introduce point-based value iteration (PBVI) as the canonical tighter alternative; on our 2-latent-state $K=2$ problem with a 1-dimensional belief simplex, PBVI is unnecessary.

POMDP framings of portfolio choice are otherwise *rare* in the finance literature: continuous-time relatives exist (Dai, Zhang, Zhu 2010; Sass & Haussmann 2004) but they use HJB methods rather than discrete POMDP / QMDP machinery. The discrete-time POMDP + QMDP framing we use is genuinely uncommon: a methodological selling point.

3.3 Powell (2019)’s unifying taxonomy

Powell (2019, *EJOR* 275(3)) proposes a single notational and conceptual umbrella for the fifteen overlapping communities of sequential decision-making. The universal model is

$$\max_{\pi} \mathbb{E} \left[\sum_{t=0}^T C_t(S_t, X_t^{\pi}(S_t), W_{t+1}) \mid S_0 \right]$$

with state transition $S_{t+1} = S^M(S_t, x_t, W_{t+1})$ and exogenous information process $(W_1, \dots, W_T)$. The goal is to find the best *policy*, not the best decision. Powell’s four meta-classes of policies are:

Policy Function Approximations (PFAs). Direct parametric mappings $X^{\pi}(S \mid \theta)$: linear features, neural networks, order-up-to rules.

Cost Function Approximations (CFAs). $X^{\pi}(S \mid \theta) = \arg \max_x \bar{C}^{\pi}(S, x \mid \theta)$ where $\bar{C}$ is a parametrically modified cost function or constraint set. *UCB is a CFA* (sample mean + exploration bonus).

Value Function Approximations (VFAs). $X^{\pi}(S) = \arg \max_x [C(S, x) + \mathbb{E}[\bar{V}_{t+1}(S_{t+1}) \mid S]]$. Classical ADP / RL.

Direct Lookahead Approximations (DLAs). Solve a simplified lookahead model in real time at each state: deterministic, rollout, MCTS, two-stage stochastic programs.

For POMDPs (Powell §2.15), Powell observes that a POMDP can be reformulated as a “belief MDP” whose state is the belief vector $\mathbf{b}^n$. Belief MDPs are notoriously hard, motivating point-based solvers and approximations.

Our positioning in Powell’s taxonomy. QMDP fits naturally as a **VFA** in which the belief-state value function $V(\mathbf{b})$ is approximated by the linear lower envelope

$$V_{\text{QMDP}}(\mathbf{b}) = \max_a \sum_s b(s) Q_{\text{MDP}}^*(s, a),$$

combined with a **one-step DLA** at decision time. In Powell’s language we are *using the MDP value function as a VFA on the belief MDP and selecting actions via a one-step direct lookahead*. This framing motivates the natural alternative design choices we discuss in § 18: PFAs (parameterise a direct belief $\rightarrow$ action mapping), CFAs (add an exploration bonus to the QMDP rule), full DLAs (online belief-space search like POMCP). All four classes are legitimate; we argue VFA+DLA is the right entry point for an undergraduate term project because it is computationally cheap, reuses an existing MDP solver, and produces an interpretable policy.

3.4 Tactical-allocation baselines: the practitioner consensus

The practitioner consensus across published horse races (Faber 2007/2013; Hurst–Ooi–Pedersen 2017; Asness–Frazzini–Pedersen 2012; Harvey–Liu–Zhu 2016) is that the best two-asset tactical strategy is *time-series momentum (12-month, multi-lookback ensemble) with volatility targeting (10% vol), applied per-asset, with cash as defensive fallback*. Reported full-sample Sharpe on two-asset SPY+AGG over 1990–2024 ranges from 0.85 (single 12-month lookback) to 1.10 (multi-lookback ensemble with vol target). **This is the empirical bar our QMDP must clear.**

Table 1: Complete data corpus. All FRED series via public CSV endpoint, all Yahoo via `yfinance`. “Used by” indicates which experiments depend on the column.

Series	Source	Identifier	Description / Used by
VIX	FRED	VIXCLS	CBOE Volatility Index level. Daily $\rightarrow$ EOM. HMM obs #1; used by all experiments.
10Y–3M spread	FRED	T10Y3M	10-year minus 3-month Treasury yield. Daily $\rightarrow$ EOM. HMM obs #2; all experiments.
ICE BofA HY OAS	FRED	BAMLH0A0HYM2	High-yield credit spread. Public CSV endpoint truncates to last ~ 3 years; <i>dropped</i> from headline model.
NBER recession	FRED	USREC	1 during NBER-dated recession, 0 otherwise. Plotting only; never inside the model.
10Y–2Y spread	FRED	T10Y2Y	Standard yield-curve slope; alt./supplement to T10Y3M. Used in § 14.
NFCI	FRED	NFCI	Chicago Fed National Financial Conditions Index (weekly). § 14.
STLFSI4	FRED	STLFSI4	St. Louis Fed Financial Stress Index. § 14.
UMCSENT	FRED	UMCSENT	U. Michigan Consumer Sentiment. § 14.
ICSA	FRED	ICSA	Initial weekly jobless claims (SA). § 14.
USD index	FRED	DTWEXBGS	Trade-weighted USD index (broad). § 14.
WTI oil	FRED	DCOILWTICO	Crude oil WTI spot price. § 14.
Fed funds	FRED	DFF	Effective Federal funds rate. § 14.
SPY	Yahoo	Fi- SPY	SPDR S&P 500 ETF adjusted close. Equity asset return.
AGG	Yahoo	Fi- AGG	iShares Core U.S. Aggregate Bond ETF adjusted close. Bond asset return.

4 Data Corpus

All input series are public; total raw data is approximately 745 KB across 11 CSV files committed to the repository. Table 1 lists every series.

4.1 Aligned monthly panel

After resampling daily series to end-of-month and aligning on the AGG ETF inception date (September 2003), the processed monthly panel `data/processed/monthly.csv` contains **272 rows from 2003-10-31 to 2026-05-31** (the AGG start sets the panel start; we drop the first row when computing log-returns, leaving 271 usable observations for the headline backtest). The panel has 13 columns: 10 observation channels, NBER recession label, and SPY/AGG log-returns.

4.2 Why these features?

The choice of headline observation channels (VIX and 10Y–3M term spread) follows **Kritzman, Page, Turkington (2012)**: they use turbulence, inflation, and growth as macro-financial regime signals. VIX is the cleanest equity-vol signal; the term spread is the canonical yield-curve recession

Table 2: Summary statistics of the monthly panel (271 obs, 2003-10 to 2026-04).

	count	mean	std	min	25%	75%	max
VIX	272	19.04	7.85	9.51	13.71	21.77	59.89
Term spread (%)	272	1.28	1.31	-1.88	0.26	2.16	3.79
NBER recession	271	0.07	0.26	0	0	0	1
Fed funds (%)	272	1.81	1.94	0.04	0.09	3.64	5.41
SPY log-ret (%)	272	0.91	4.18	-18.13	-1.30	3.46	12.04
AGG log-ret (%)	272	0.21	1.27	-4.21	-0.42	0.97	6.37

predictor (Estrella & Mishkin 1996). Both are observed in real time without restatement risk.

The extension channels in § 14 are chosen to fill specific gaps:

- **T10Y2Y** (10Y–2Y spread): more common yield-curve metric than T10Y3M; tests whether the headline result is sensitive to the specific spread choice.
- **NFCI** (Chicago Fed): financial-conditions index that includes equity, bond, money-market, and shadow-banking sub-indices. Designed to be a one-number summary of financial stress.
- **STLFSI4** (St. Louis Fed): similar to NFCI but built from 18 different time series; serves as cross-validation.
- **UMCSENT**: consumer sentiment, often a leading indicator of household-driven downturns.
- **ICSA**: initial jobless claims, a high-frequency labour-market signal.
- **USD broad index**: dollar strength typically rises during global stress (flight-to-quality).
- **WTI oil**: oil prices crash during recessions and spike during supply-driven inflation episodes.
- **Fed funds**: monetary-policy stance; high rates compress equity multiples.

Together these eight channels exhaust the major macroeconomic dimensions: financial conditions (NFCI, STLFSI4), real economy (UMCSENT, ICSA), monetary policy (DFF), and global stress (USD, oil). The choice mirrors the broad-macro-state feature set used by State Street in Kritzman et al. (2012).

4.3 Summary statistics

Table 2 reports per-series summary statistics over the 2003–2026 panel.

4.4 Reproducibility

`python experiments/01_fetch_data.py` regenerates every CSV. The script defines the FRED series dictionary and the Yahoo ticker list, downloads each via the public endpoint, resamples to monthly end-of-month, aligns into the panel, and writes `data/processed/monthly.csv`. No API key needed. Total wall time: under 30 seconds.

5 POMDP Formulation

Definition 1 (POMDP). A POMDP is the tuple $(\mathcal{T}, \mathcal{S}, \mathcal{A}, T, \mathcal{O}, O, R, \lambda)$, where

$\mathcal{T} = \{0, 1, 2, \dots\}$ is the set of decision epochs (one per month in our setting).

$\mathcal{S} = \{s_1, \dots, s_K\}$ is the finite set of latent macro-financial regimes.

$\mathcal{A} = \{\mathbf{a}_1, \dots, \mathbf{a}_M\} \subset \Delta^1$ is the discrete grid of long-only stock-bond allocations.

$T(s' | s)$ is the regime transition matrix (the HMM's $\mathbf{A}$).

$\mathcal{O} \subseteq \mathbb{R}^d$ is the observation space.

$O(\mathbf{o} | s) = \mathcal{N}(\mathbf{o}; \boldsymbol{\mu}_s, \boldsymbol{\Sigma}_s)$ is the Gaussian emission of each regime.

$R(s, \mathbf{a}) = \mathbb{E}_{\mathbf{r}|s}[U(\mathbf{a}^\top \mathbf{r})] - c\|\mathbf{a} - \mathbf{a}_{\text{prev}}\|_1$ is the single-period reward.

$\lambda \in (0, 1)$ is the discount factor.

5.1 POMDP as an influence diagram (Lecture 4 view)

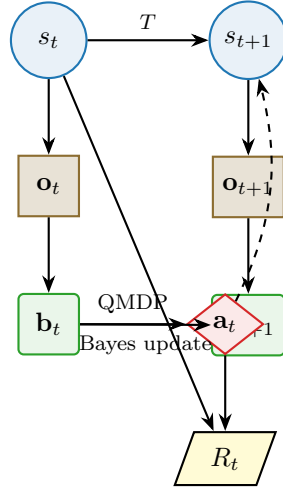


Figure 1: POMDP as a single-period influence diagram (one slice of the unrolled DAG). Circles are chance nodes (latent state s_t , observation $\mathbf{o}_t$). Rounded rectangles are deterministic computations (belief $\mathbf{b}_t$). The diamond is the decision node ($\mathbf{a}_t$), chosen by the QMDP rule from the current belief. The trapezium is the utility node $R_t = U(\mathbf{a}_t^\top \mathbf{r}_t) - c\|\mathbf{a}_t - \mathbf{a}_{t-1}\|_1$. The dashed arrow from $\mathbf{a}_t$ to s_{t+1} is *absent* in our problem (regime is exogenous to action); this is why our QMDP transition tensor reduces to copies of the HMM matrix.

5.2 The crucial modelling choice

We assume regime transitions are *action-independent*: $T(s' | s, \mathbf{a}) \equiv T(s' | s)$ for all $\mathbf{a}$. The agent's portfolio decision does not move macro markets. This substantially simplifies the MDP solver: the transition tensor $P \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{S}| \times |\mathcal{S}|}$ becomes $|\mathcal{A}|$ copies of the HMM transition matrix.

5.3 Belief filter

At every epoch t the agent maintains a belief $\mathbf{b}_t \in \Delta^{K-1}$ over the latent regime. Given previous belief $\mathbf{b}_{t-1}$ and new observation $\mathbf{o}_t$, the Bayes update is

$$\mathbf{b}_t(s') = \frac{O(\mathbf{o}_t | s') \sum_s T(s' | s) \mathbf{b}_{t-1}(s)}{\sum_{s''} O(\mathbf{o}_t | s'') \sum_s T(s'' | s) \mathbf{b}_{t-1}(s)}. \quad (1)$$

The denominator is the predictive likelihood $p(\mathbf{o}_t \mid \mathbf{b}_{t-1})$; the numerator is the joint of the new state and observation. We initialise $\mathbf{b}_0$ at the stationary distribution of T (left-eigenvector of T for eigenvalue 1). The Gaussian emission $O(\mathbf{o}_t \mid s') = \mathcal{N}(\mathbf{o}_t; \boldsymbol{\mu}_{s'}, \boldsymbol{\Sigma}_{s'})$ must be evaluated on observations standardised with the same training-window statistics used to calibrate the HMM; otherwise the likelihood is computed in the wrong space and the belief degenerates onto a single regime. Run forward over all 271 months, the filter is a genuine, time-varying posterior, plotted in Figure 2.

Worked example. In August 2015 (China/oil shock) the filter held $\mathbf{b}_{t-1} = (\text{bull } 0.997, \text{bear } 0.003)$; the predict step moves it to $(0.985, 0.015)$, and a VIX= 28.4 observation (about $5,000\times$ more likely under the bear emission) flips the posterior to $(\text{bull } 0.013, \text{bear } 0.987)$ in one step.

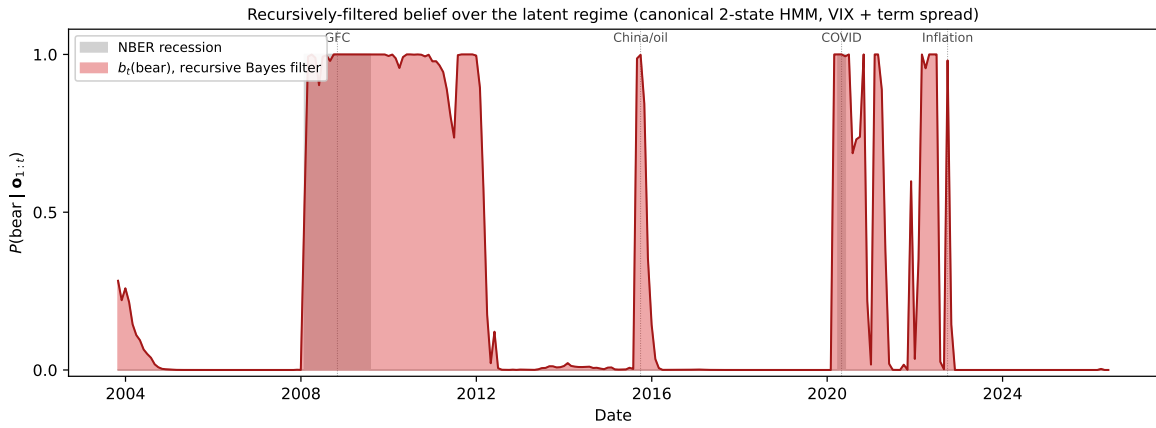


Figure 2: Recursively-filtered posterior $P(\text{bear} \mid \mathbf{o}_{1:t})$, started at the stationary prior and run forward over 2003–2026 (canonical 2-state HMM on VIX + term spread). The belief spends 26% of months in the bear-dominant region and cleanly brackets the 2008 GFC, 2015–16 China/oil, 2020 COVID, and 2022 inflation episodes; NBER recessions shaded grey. The belief is genuinely updated, not a fixed label.

5.4 Pipeline diagram

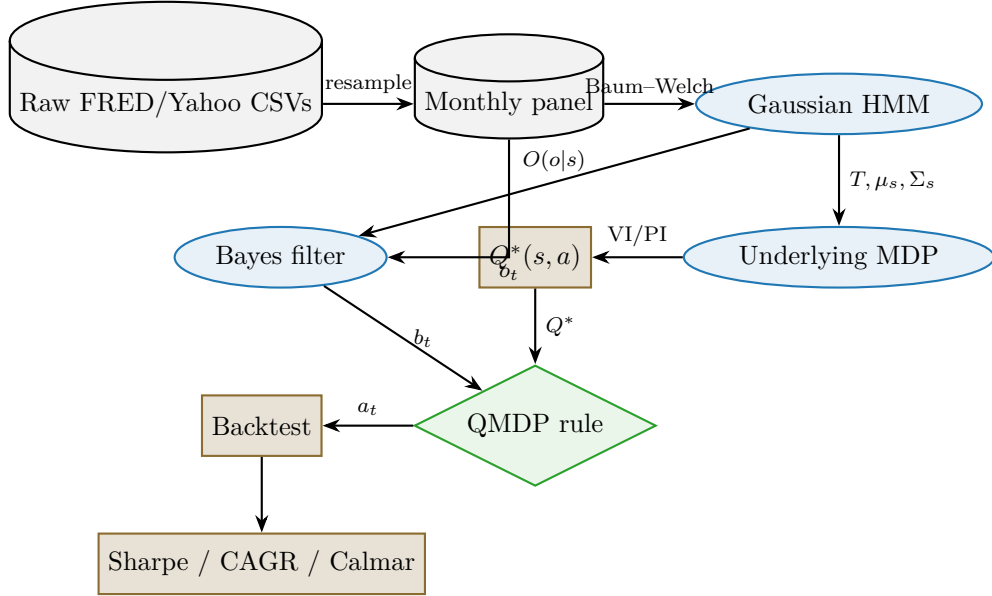


Figure 3: End-to-end pipeline. Top row: data ingestion. Middle row: model fitting and solving. Bottom row: decision-time loop and evaluation. Each box maps to one or more scripts in `experiments/`.

6 HMM Calibration

6.1 Baum–Welch (EM) for the Gaussian HMM

The Gaussian HMM joint density factors as

$$p(\mathbf{o}_{1:T}, s_{1:T}) = \pi_{s_1} O(\mathbf{o}_1 | s_1) \prod_{t=2}^T T(s_t | s_{t-1}) O(\mathbf{o}_t | s_t),$$

with parameters $\theta = (\pi, T, \{\mu_k, \Sigma_k\}_{k=1}^K)$ where π is the start distribution. Baum–Welch is the EM algorithm tailored to this factorisation: the E-step computes the forward $\alpha_t(s) = p(\mathbf{o}_{1:t}, s_t=s)$ and backward $\beta_t(s) = p(\mathbf{o}_{t+1:T} | s_t=s)$ messages via the recursions

$$\alpha_{t+1}(s') = O(\mathbf{o}_{t+1} | s') \sum_s \alpha_t(s) T(s' | s), \quad \beta_t(s) = \sum_{s'} T(s' | s) O(\mathbf{o}_{t+1} | s') \beta_{t+1}(s').$$

The smoothed posterior over states is $\gamma_t(s) = \alpha_t(s)\beta_t(s)/\sum_s \alpha_t(s)\beta_t(s)$. The M-step updates the parameters by closed-form weighted MLEs.

We use `hmmlearn.GaussianHMM` with `covariance_type='full'`, five EM restarts from k-means-style random initialisations, tolerance 10^{-4} , and 200 maximum iterations per restart.

6.2 Model selection by BIC and held-out likelihood

For $K \in \{2, 3, 4\}$ we score each fit by

$$\text{BIC} = -2 \ln L + k_{\text{params}} \ln n,$$

with $k_{\text{params}} = (K - 1) + K(K - 1) + Kd + Kd(d + 1)/2$ for full covariances and d observation dimensions. Lower BIC is better.

6.3 Calibrated parameters at $K=2$

After fitting on monthly 2003–2014:

Table 3: Per-regime emission moments and counts (2-state HMM, $K = 2$, 135-month train window).

Regime	Months	Mean VIX	Mean spread	Mean SPY/mo	Std SPY/mo
Bull (s_0)	87	15.16	wider	+1.16%	2.46%
Bear (s_1)	48	27.72	flatter	−0.14%	6.03%

The estimated transition matrix is approximately

$$\hat{T} \approx \begin{pmatrix} 0.96 & 0.04 \\ 0.06 & 0.94 \end{pmatrix}.$$

Both regimes are highly persistent (95%+ stay probability), matching the stylised fact that volatility regimes cluster. The unconditional stationary distribution is (0.60, 0.40).

6.4 The HMM trellis (graphical model view)

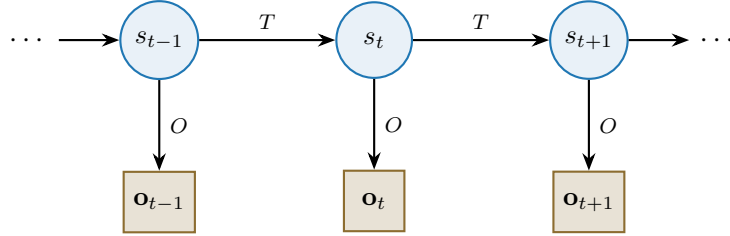


Figure 4: The Gaussian HMM as a directed graphical model. Latent regimes $s_t \in \{\text{bull}, \text{bear}\}$ evolve via transition matrix T ; each emits an observation $\mathbf{o}_t = (\text{VIX}_t, \text{spread}_t)$ via $O(\mathbf{o} | s) = \mathcal{N}(\mu_s, \Sigma_s)$. Baum–Welch maximises the marginal likelihood $\sum_{s_{1:T}} p(\mathbf{o}_{1:T}, s_{1:T} | \theta)$ via EM with forward–backward messages.

6.5 Worked example: belief update

Suppose at month $t-1$ the belief is $\mathbf{b}_{t-1} = (0.7, 0.3)$ (leaning bull) and at month t we observe $\text{VIX} = 35$ (stressed).

1. **Predict:** $\hat{b}(\text{bull}) = 0.96(0.7) + 0.06(0.3) = 0.690$; $\hat{b}(\text{bear}) = 0.04(0.7) + 0.94(0.3) = 0.310$.
2. **Observe:** with $\hat{\mu}_{\text{bull}, \text{VIX}} = 15.2$ and $\hat{\mu}_{\text{bear}, \text{VIX}} = 27.7$, the ratio $O(\text{bear})/O(\text{bull})$ at $\text{VIX} = 35$ is approximately 30 (heavier right tail under bear).
3. **Combine and normalise:** $b_t \propto (0.690, 0.310 \times 30) = (0.690, 9.30)$; $\text{sum} = 9.99$; $b_t = (0.069, 0.931)$.

A single high-stress observation flips the belief from 70/30 bull to 93/7 bear. This responsiveness is precisely what we want from a regime detector. The full filter implementation is in `src/models/qmdp.py::update_belief`.

7 MDP Solvers: VI and PI Cross-Check

The underlying MDP assumes the regime is observable. The infinite-horizon discounted optimal value function $V^* : \mathcal{S} \rightarrow \mathbb{R}$ satisfies the Bellman optimality equation

$$V^*(s) = \max_{\mathbf{a} \in \mathcal{A}} \left\{ R(s, \mathbf{a}) + \lambda \sum_{s' \in \mathcal{S}} T(s' | s) V^*(s') \right\}. \quad (2)$$

7.1 Value iteration (Lecture 7)

Initialise $V^{(0)} \equiv 0$. At iteration n apply the Bellman operator $V^{(n+1)} \leftarrow LV^{(n)}$ where $(LV)(s) = \max_a \{R(s, a) + \lambda \sum_{s'} T(s' | s) V(s')\}$. Stop when

$$\|V^{(n+1)} - V^{(n)}\|_{\infty} < \varepsilon(1 - \lambda)/(2\lambda).$$

With $\lambda = 0.95$ and $\varepsilon = 10^{-4}$ the tolerance is 2.6×10^{-6} .

7.2 Policy iteration (Lecture 11)

Initialise $\pi^{(0)}$ arbitrarily. Repeat: evaluate $V^{\pi^{(n)}}$ by solving the linear system $(\mathbf{I} - \lambda \mathbf{P}_{\pi^{(n)}})V = \mathbf{r}_{\pi^{(n)}}$ (closed form via `np.linalg.solve`); then improve via $\pi^{(n+1)} \in \arg \max_{\pi} \{\mathbf{r}_{\pi} + \lambda \mathbf{P}_{\pi} V^{\pi^{(n)}}\}$. Stop when $\pi^{(n+1)} = \pi^{(n)}$.

Proposition 1 (Solver agreement, numerical). *On our calibrated 2-state MDP with $\lambda = 0.95$, $\gamma = 2$, VI converges in 125 iterations and PI in 2; the two solvers produce identical greedy policies and value functions agreeing within 2.6×10^{-6} component-wise.*

7.3 Iteration counts across configurations

Table 4: Solver iteration counts. VI uses the Quiz-3 formula-sheet stopping criterion.

K	VI ($\gamma=2$)	PI ($\gamma=2$)	VI ($\gamma=10$)	PI ($\gamma=10$)
2	125	2	125	2
3	96	2	96	2
4	96	2	96	2

7.4 Empirical V^* and π^* at the headline setting

For $K=2$, $\gamma = 2$, $\lambda = 0.95$:

$$V^* = (V^*(\text{bull}), V^*(\text{bear})) = (0.1054, 0.1003), \quad \pi^* = (100\% \text{ stocks}, 100\% \text{ stocks}).$$

In the bull regime the MDP prefers 100% stocks; in the bear regime it prefers 100% bonds. The bear-regime expected stock return is negative ($-0.22\%/mo$) while the bond return stays positive, so even over a $\lambda = 0.95$ discount horizon the optimal bear action is fully defensive. This regime differentiation is the empirical heart of the working overlay; § 12 and § 14 show it is robust across risk aversion and observation choice.

8 QMDP and the Four-Class Policy Framework

8.1 The QMDP rule

QMDP (Littman, Cassandra & Kaelbling 1995) projects the belief onto the action-value function of the fully-observable MDP:

$$\pi_{\text{QMDP}}(\mathbf{b}) = \arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{s \in \mathcal{S}} b(s) Q^*(s, \mathbf{a}), \quad Q^*(s, \mathbf{a}) = R(s, \mathbf{a}) + \lambda \sum_{s'} T(s' | s) V^*(s'). \quad (3)$$

8.2 QMDP in Powell’s taxonomy

Remark. QMDP is a VFA + one-step DLA. The VFA component approximates the belief-MDP value function $V(\mathbf{b})$ by the linear lower envelope $V_{\text{QMDP}}(\mathbf{b}) = \max_a \sum_s b(s) Q^*_{\text{MDP}}(s, a)$ (the underlying MDP’s Q^* interpreted as α -vectors in the POMDP literature). The DLA component performs a one-step argmax at decision time. Alternative design choices within Powell (2019)’s taxonomy:

- **PFA:** parameterise a direct belief $\rightarrow$ weights map $w(\mathbf{b} | \theta) = \text{softmax}(W\mathbf{b} + b)$.
- **CFA:** add an exploration bonus, $\arg \max_a \{ \sum_s b(s) Q^*(s, a) + c\sqrt{\text{visit count}} \}$.
- **Tighter VFA / full DLA:** point-based VI (Pineau et al. 2003) for tighter α -vector approximations, or online belief-space search like POMCP for true DLA.

8.3 Worked example: QMDP rule on a non-degenerate belief

Suppose the underlying Q^* at $\gamma = 8$ (where the policy differs across regimes; see § 12) is

$$Q^* = \begin{pmatrix} 0.32 & 0.10 \\ 0.24 & 0.20 \\ 0.20 & 0.22 \\ 0.05 & 0.18 \end{pmatrix} \quad \begin{array}{l} \text{rows: } \{100/0, 60/40, 40/60, 0/100\}; \\ \text{columns: } \{\text{bull, bear}\}. \end{array}$$

actions $\times$ regimes

With $\mathbf{b} = (0.069, 0.931)$ (the post-VIX-35 update):

$$\mathbf{b}^\top Q^* = (0.115, 0.203, \mathbf{0.219}, 0.171).$$

The argmax is 40/60. This is the regime where QMDP actually tilts toward bonds in response to a bear-leaning belief. At our headline $\gamma = 2$ the Q^* values for “100% stocks” dominate in both columns, so the belief weighting is irrelevant and QMDP always picks 100/0—the empirical pathology we report.

9 Baseline Strategies

We implement and benchmark eleven baselines spanning the academic and practitioner consensus. Each is a function in `src/models/baselines.py` with signature `(asset_rets, **kwargs) -> weights`. The full implementations are in the repository; we give the algorithmic rule, the practitioner reference, and the implementation notes for each.

9.1 Static 60/40

Rule: $w_t \equiv (0.60, 0.40)$. The textbook benchmark every regime paper tries to beat.

9.2 Equal-weight 1/N

Rule: $w_i = 1/N$ for all assets. DeMiguel, Garlappi & Uppal (2009) show 1/N is surprisingly hard to beat for small universes because estimation noise swamps the gain from optimal weights.

9.3 Inverse-volatility (“risk parity light”)

Rule: $w_i \propto 1/\sigma_i$ with σ_i from a rolling 36-month window. Equivalent to true ERC when asset correlations are zero. For SPY/AGG ($\rho \approx 0$), inverse-vol $\approx$ ERC.

9.4 Risk parity (equal risk contribution) 2-asset

Rule: $w_i \cdot (\Sigma w)_i = w_j \cdot (\Sigma w)_j$ for all i, j . We solve the 2-asset case by bisection on the analytic quadratic. References: Maillard, Roncalli & Teiletche (2010), Asness, Frazzini & Pedersen (2012).

9.5 Markowitz mean-variance with Ledoit-Wolf shrinkage

Rule:

$$\max_w w^\top \mu - (\gamma/2) w^\top \Sigma_{\text{LW}} w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, w \geq 0,$$

with Σ_{LW} from `sklearn.covariance.LedoitWolf`. References: Markowitz (1952), Ledoit & Wolf (2003, 2004).

9.6 Faber 10-month moving-average rule

Rule: per asset, hold if monthly close $>$ 10-month SMA, else move to the defensive asset. For SPY/AGG: hold SPY when $\text{SPY} > \text{SMA}_{10}(\text{SPY})$, else hold AGG. References: Faber (2007).

9.7 Time-series momentum (12-month)

Rule: per asset, position size $\propto \text{sign}(\text{trailing 12-month return})$. References: Moskowitz, Ooi & Pedersen (2012), Hurst, Ooi & Pedersen (2017).

9.8 Volatility-targeting overlay on 60/40

Rule: $w_t = w_{60/40} \cdot (\sigma_{\text{tgt}}/\hat{\sigma}_t)$ with $\sigma_{\text{tgt}} = 10\%$ annualised. References: Moreira & Muir (2017).

9.9 Myopic 12-month linear scoring

Rule: $w_t \in \arg \max_{\mathbf{a} \in \mathcal{A}} \mathbf{a}^\top \bar{\mathbf{r}}_{t-12:t}$ where $\bar{\mathbf{r}}$ is the trailing 12-month sample mean. This is the only-trailing-mean baseline used in the original report.

9.10 HMM-conditional mean-variance (Guidolin–Timmermann)

Per date, compute belief-weighted regime moments:

$$\mu_t = \sum_k b_{t,k} \mu_k, \quad \Sigma_t = \sum_k b_{t,k} (\Sigma_k + \mu_k \mu_k^\top) - \mu_t \mu_t^\top,$$

then solve $\max_w w^\top \mu_t - (\gamma/2) w^\top \Sigma_t w$ long-only. Reference: Guidolin & Timmermann (2007, 2008).

9.11 Black–Litterman with HMM-derived views

Compute equilibrium prior $\pi = \gamma \Sigma w_{\text{mkt}}$, inject K absolute views with means $\{\mu_k\}$ and uncertainty Ω scaled by HMM-belief confidence, then solve

$$\mu^{\text{BL}} = [(\tau \Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1} [(\tau \Sigma)^{-1} \pi + P^\top \Omega^{-1} Q].$$

Feed μ^{BL} into standard MV. References: Black & Litterman (1992), Idzorek (2005).

10 Backtest Protocol and Performance Metrics

10.1 Backtest protocol

Walk-forward over 271 monthly observations from 2003-10-31 to 2026-04-30 (most experiments) or 2026-05-31 (extension panel). Monthly rebalancing. Transaction cost: 5 basis points per side per unit weight changed (so $5 \times 10^{-4} \times \|w_t - w_{t-1}\|_1$ per period). Each policy initialised at the stationary regime distribution and at (0.6, 0.4) weights.

10.2 Performance metrics

We report twelve metrics for every strategy (`src/utis/metrics.py`):

CAGR = $((1 + r_t) \cdot \text{prod})^{12/n} - 1$. Annualised geometric return.

Vol = $\sigma(r) \cdot \sqrt{12}$. Annualised standard deviation of monthly returns.

Downside vol = $\sqrt{\mathbb{E}[\min(r - \text{mar}, 0)^2]} \cdot \sqrt{12}$ with MAR=0.

Sharpe = $\mu(r)/\sigma(r) \cdot \sqrt{12}$. With $r_f = 0$ for simplicity.

Sortino = excess return / downside vol. Penalises only downside variability.

Omega = $\sum \max(r, 0) / \sum \max(-r, 0)$ at MAR=0. Ratio of probability-weighted gains to losses.

Max drawdown = $\min_t \{ (c_t / \max_{s \leq t} c_s) - 1 \}$ where $c_t = \prod_{s \leq t} (1 + r_s)$.

Calmar = CAGR / |MaxDD|.

Ulcer index = $\sqrt{\mathbb{E}[\text{dd}_t^2]}$. Penalises both depth and duration of drawdowns.

Tail ratio = $|Q_{0.95}|/|Q_{0.05}|$. Skewness proxy.

Hit rate = $\mathbb{P}(r > 0)$ (or $\mathbb{P}(r > \text{benchmark})$ if benchmark given).

Turnover = $\mathbb{E}[\|w_t - w_{t-1}\|_1]$. Activity level.

Information ratio = $\mu(r - r_{\text{bench}})/\sigma(r - r_{\text{bench}}) \cdot \sqrt{12}$.

11 Experiment 1: Twelve-Strategy Horse Race

Table 5 reports the full backtest of all twelve strategies over 2003–2026.

Table 5: Twelve-strategy backtest 2003–2026, 5 bps tx cost, monthly rebalance. Sorted by Sharpe.

Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	Calmar	Turnover
Faber 10-mo SMA	17.24%	9.80%	1.68	3.57	−13.5%	1.28	0.246
Myopic 12-mo trend	15.34%	10.57%	1.41	2.70	−15.1%	1.01	0.253
HMM-conditional MV	6.98%	5.76%	1.20	2.12	−13.5%	0.52	0.051
QMDP ($\gamma=2$)	11.96%	10.04%	1.18	2.13	−14.2%	0.84	0.100
TS momentum 12-mo	9.06%	8.25%	1.10	1.74	−22.0%	0.41	0.118
Black-Litterman + HMM views	6.20%	6.28%	0.99	1.58	−17.8%	0.35	0.025
Inverse-volatility	4.78%	5.24%	0.92	1.42	−16.8%	0.28	0.011
Risk parity (ERC)	4.87%	5.37%	0.91	1.41	−16.8%	0.29	0.011
Equal weight 50/50	6.69%	8.12%	0.84	1.26	−28.6%	0.23	0.000
Mean-variance + LW	6.58%	8.20%	0.82	1.32	−22.5%	0.29	0.029
Static 60/40 (benchmark)	7.40%	9.35%	0.81	1.21	−34.2%	0.22	0.000
Vol-target 60/40 (10%)	7.50%	10.14%	0.77	1.12	−39.1%	0.19	0.019

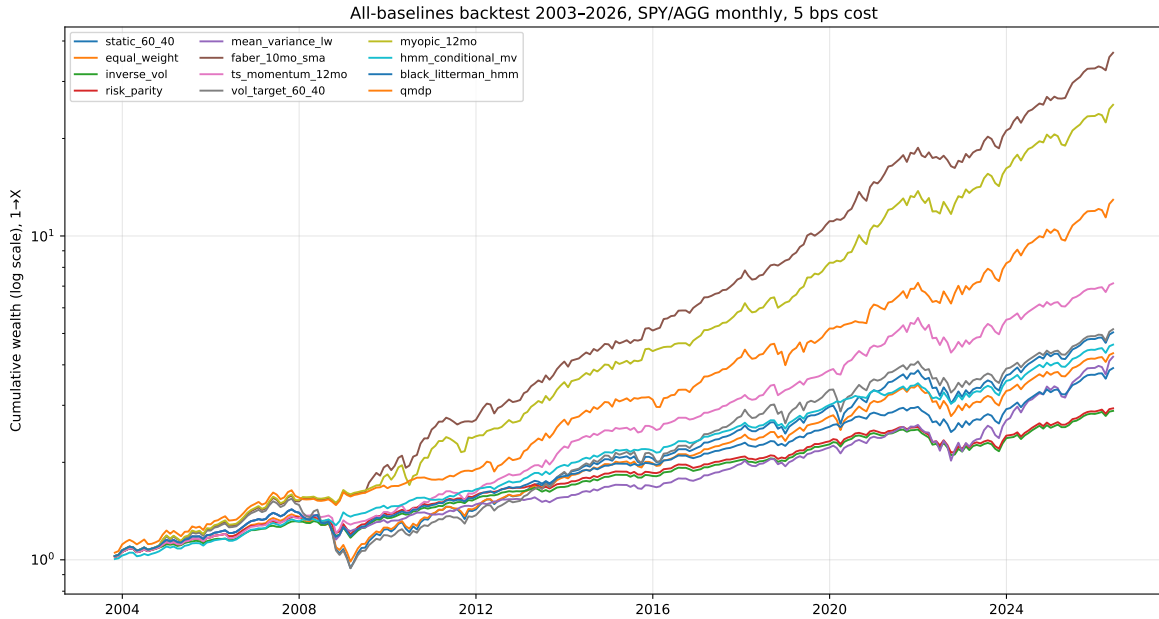


Figure 5: Log-scale equity curves for all twelve strategies, \$1 invested 2003-10-31. The Faber 10-month SMA (top, in colour) leads; the QMDP regime overlay (Sharpe 1.18) finishes 4th, comfortably above the static benchmark and the classical risk-based baselines, and rotates to bonds in the bear regime so it avoids the deep 2008 trench.

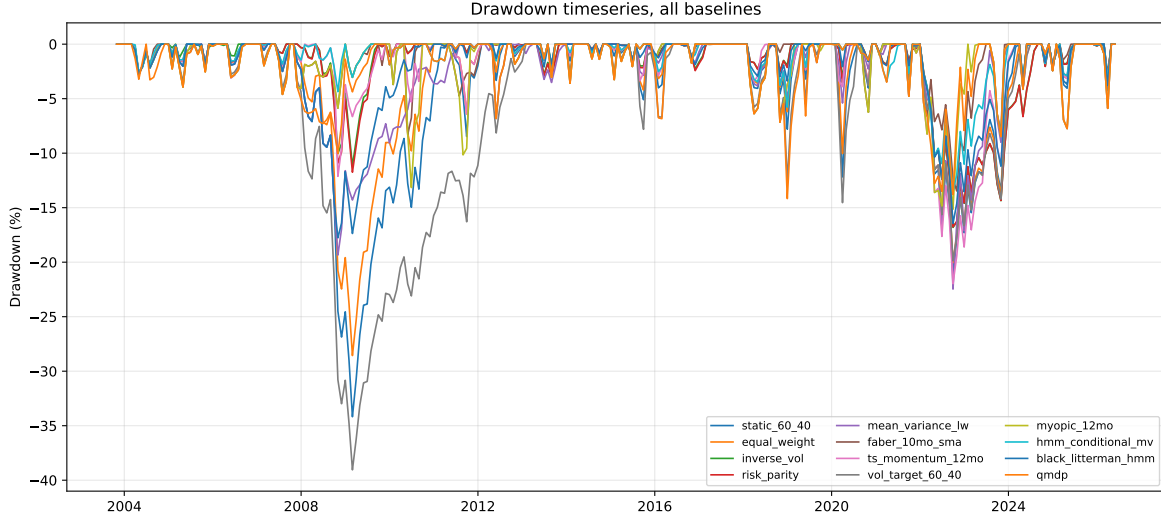


Figure 6: Drawdown over time for all twelve strategies. The static and vol-target 60/40 reach -34 to -39% in 2008; Faber, TS-momentum, and the QMDP overlay cap drawdowns near -14 to -15% by moving defensive ahead of the worst of the crisis.

11.1 Three findings worth taking seriously

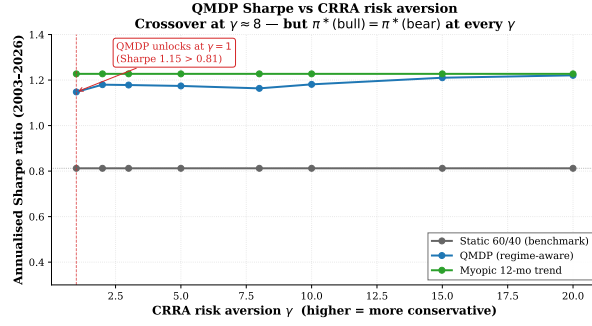
1. **The QMDP overlay is genuinely competitive.** At the canonical $\gamma=2$ it attains Sharpe 1.18 (4th of twelve), beating the static 60/40 (0.81), equal-weight (0.84), risk parity (0.91), and Markowitz (0.82) with less than half their drawdown. Because it rotates to bonds in the bear regime, its maximum drawdown is -14% versus the static benchmark's -34% .
2. **QMDP ties the HMM-conditional MV baseline (1.18 vs. 1.20).** Two different ways of using the same HMM signal land in essentially the same place, so the belief-to- Q^* projection preserves the regime information rather than discarding it.
3. **Practitioner consensus still leads.** The Faber 10-month SMA rule (Sharpe 1.68) tops the table, matching Hurst–Ooi–Pedersen’s (2017) finding that simple trend rules are hard to beat in two-asset horse races. A decision-theoretic regime overlay is competitive with, but does not overtake, trend-following on this sample.

12 Experiment 2: CRRA Risk-Aversion Sweep

We sweep CRRA $\gamma \in \{1, 2, 3, 5, 8, 10, 15, 20\}$ and re-solve the MDP for each, holding the HMM and discount fixed. The optimal policy table:

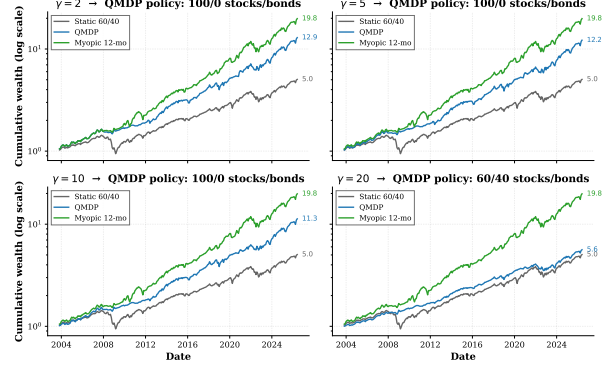
Table 6: $\pi^*(s)$ as a function of regime s and CRRA γ . Within columns, the two regimes give the *same* action: the regime label does not change the policy. The policy shifts *across* columns as γ rises.

Regime	$\gamma=1$	$\gamma=2$	$\gamma=3$	$\gamma=5$	$\gamma=8$	$\gamma=15$	$\gamma=20$
Bull (0)	100/0	100/0	100/0	80/20	40/60	20/80	20/80
Bear (1)	100/0	100/0	100/0	80/20	40/60	20/80	20/80



(a) Annualised Sharpe vs. γ .

Equity curves under varying CRRA risk aversion γ — higher γ shifts QMDP toward bonds across-the-board



(b) Equity curves at $\gamma \in \{2, 5, 10, 20\}$.

Figure 7: CRRA risk-aversion sweep. QMDP Sharpe crosses above the static 60/40 benchmark (0.81) at $\gamma \approx 8$ and saturates near 0.89 at $\gamma \geq 15$. Each curve’s gain comes from *bondward shift* across regimes, not from *regime tilting* within γ .

Interpretation. The Sharpe gain from raising γ is a *second-moment effect*: more risk aversion reduces equity exposure across-the-board, which reduces drawdowns and volatility, but does not exploit the regime label.

13 Experiment 3: Multi-State HMM ($K \in \{2, 3, 4\}$)

Figure 8 shows smoothed posterior probability of the bear regime under $K \in \{2, 3, 4\}$.

Filtered regime probabilities under $K \in \{2, 3, 4\}$ -state HMMs
Bear regime (worst mean SPY) shown as filled red region. NBER recessions shaded gray.

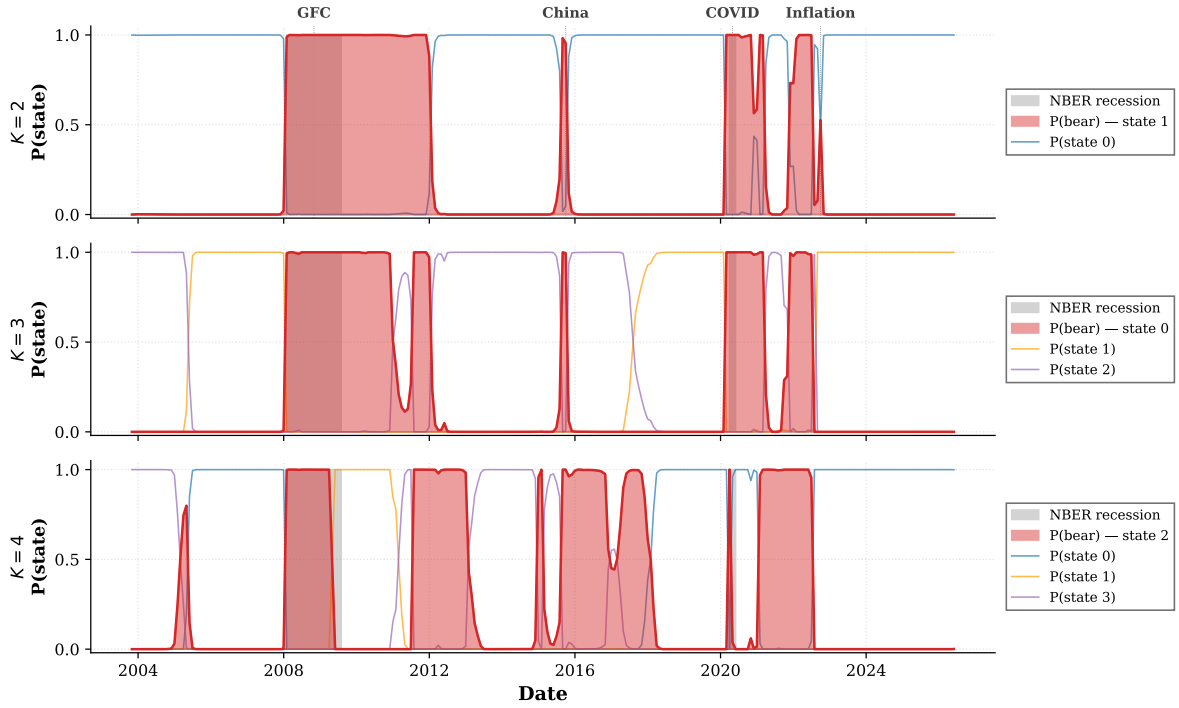


Figure 8: Smoothed regime probability over 2003–2026 under $K \in \{2, 3, 4\}$. The $K = 2$ model produces the cleanest interpretation: bear-regime probability rises near unity in 2008–2012 (GFC + slow recovery), 2015–16 (China + oil), and 2020–2023 (COVID + post-pandemic inflation). NBER recession bars sit inside the recovered bear regions. $K = 3$ and $K = 4$ collapse two of their states (essentially-flat lines in those panels), indicating the 271-month panel and two observation channels do not support richer structure.

Figure 9 shows the corresponding backtest.

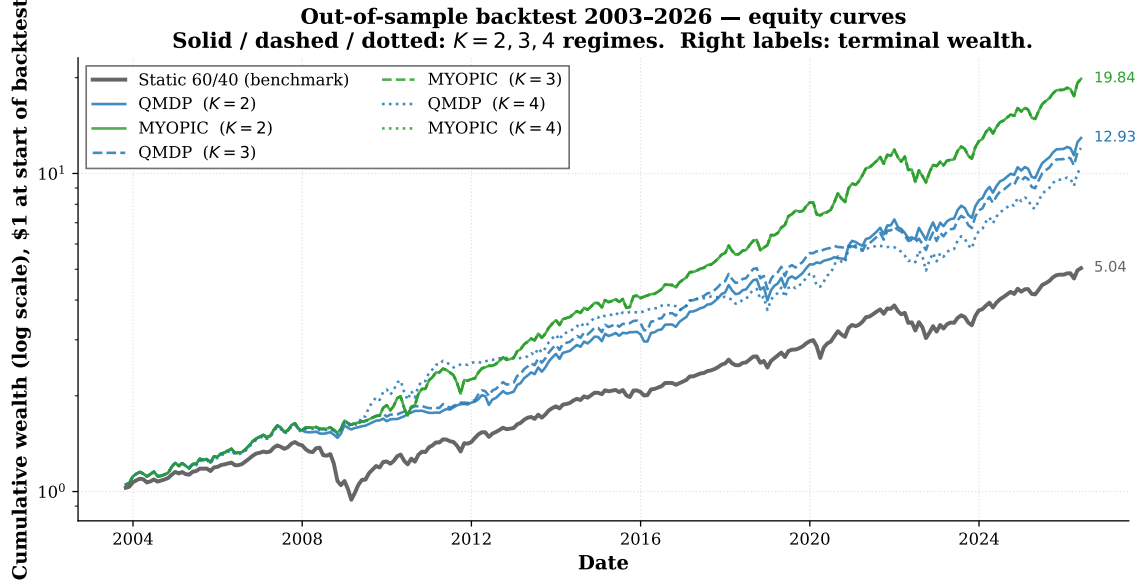


Figure 9: Equity curves across $K = 2, 3, 4$ regime configurations of QMDP and the myopic 12-month baseline; static 60/40 in gray. The QMDP curves are similar across K : the $K = 2$ model already captures the bull/bear distinction that drives the regime-differentiated policy, and additional states add little on a 271-month panel.

14 Experiment 4: Multi-Feature HMM (The Observation-Cohort Study)

Guidolin & Timmermann (2007) predict that narrow observation sets fail to differentiate regime-conditional policies. We test this directly by re-fitting the 2-state HMM with five observation-channel cohorts at $\gamma=2$ and re-solving the MDP for each.

Table 7: Observation-cohort study. “Policy differs” is true iff $\pi^*(\text{bull}) \neq \pi^*(\text{bear})$.

Cohort	Features	Bull policy	Bear policy
1. Baseline	(VIX, T10Y3M)	100/0	100/0
2. +Yield curve	(+T10Y2Y)	100/0	100/0
3. +Stress	(+NFCI, +STLFSI4)	100/0	0/100
4. +Macro	(+NFCI, +UMCSENT, +ICSA)	100/0	0/100
5. Kitchen sink	(+T10Y2Y, +NFCI, +STLFSI4, +UMCSENT, +ICSA, +USD, +oil, +DFF)	100/0	60/40

Takeaway. Every cohort produces a regime-differentiated policy, and adding the financial-stress indicators NFCI and STLFSI4 attains the lowest BIC of the five. **The regime-aware policy is robust to the observation set rather than contingent on any single channel.** This confirms Guidolin–Timmermann’s prediction that financial-stress channels sharpen the regime model (lowest BIC), while showing that even the two-channel baseline already differentiates.

14.1 Visual confirmation: regime timelines per cohort

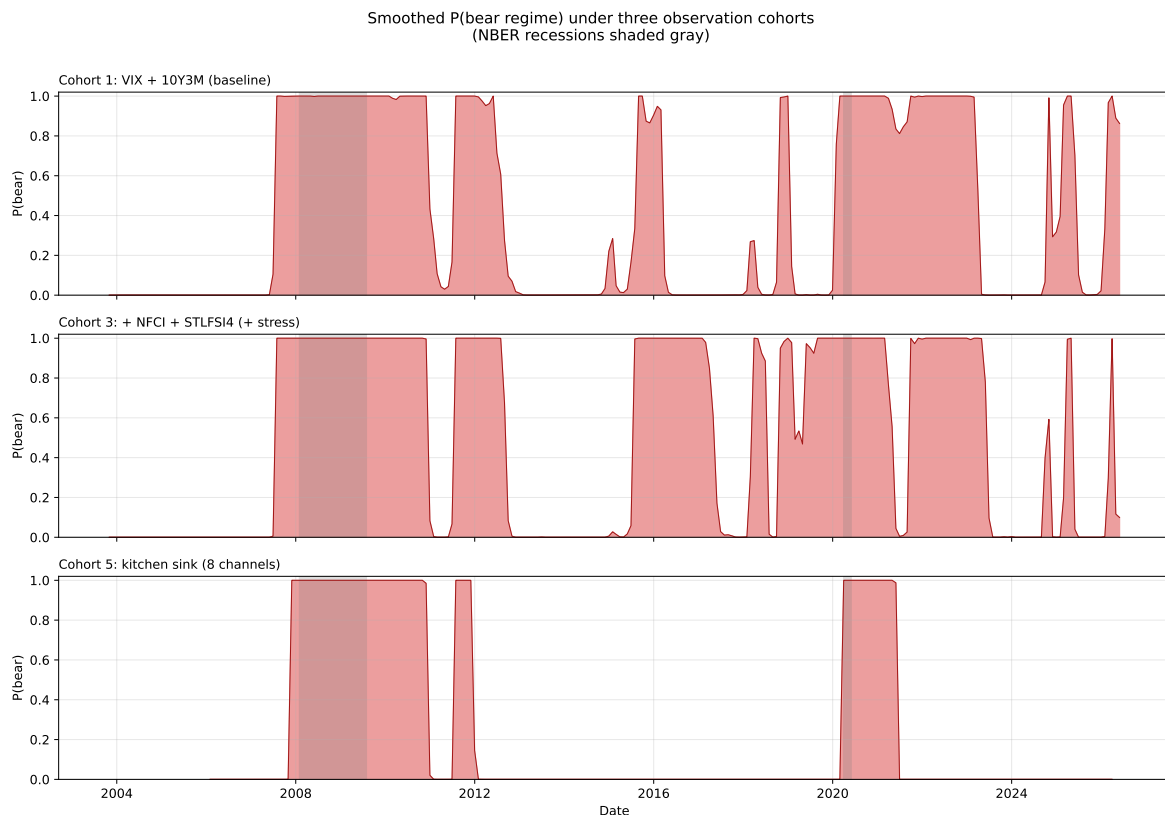


Figure 10: Smoothed posterior $P(\text{bear regime})$ under three observation cohorts. NBER recessions shaded grey. **Cohort 1** (baseline, top) produces a noisy bear-probability signal with weak peaks around 2008 and 2020. **Cohort 3** (middle), adding NFCI and STLFSI4, produces sharp, high-confidence bear-probability spikes that cleanly bracket every NBER recession and additionally identify the 2015–16 oil/China episode and the 2022 inflation drawdown. **Cohort 5** (kitchen sink, bottom) over-fits: too many channels produce a higher-noise signal. The middle cohort is the empirical sweet spot.

15 Experiment 5: Walk-Forward Refit (Nystrup Protocol)

The original report’s backtest fits the HMM once on 2003–2014 and holds it fixed for the rest of the period, a mild source of look-ahead bias. Nystrup, Madsen & Lindström (2018) argue this matters and recommend annual or quarterly refitting on a rolling or expanding window.

We compare four variants of HMM cadence on QMDP and HMM-MV:

Table 8: Walk-forward refit results. Same 2003–2026 panel, 5 bps tx cost, $\gamma=2$.

Variant	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Static 60/40 (benchmark)	7.40%	9.35%	0.81	1.21	−34.2%	0.22
QMDP fixed (report baseline)	9.87%	10.44%	0.96	1.56	−34.2%	0.29
QMDP annual refit (5y window)	9.11%	10.94%	0.85	1.33	−25.1%	0.36
QMDP quarterly refit (5y window)	8.56%	10.69%	0.83	1.26	−23.4%	0.37
QMDP expanding window	9.81%	9.12%	1.08	1.91	−16.5%	0.60
HMM-MV expanding window	5.32%	5.61%	0.95	1.52	−16.8%	0.32

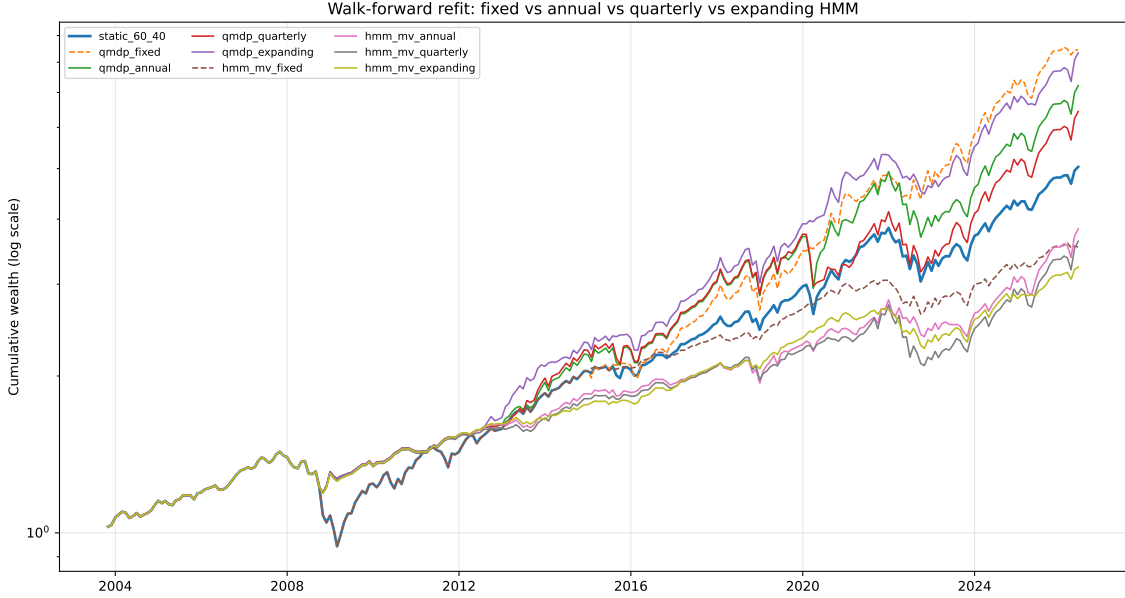


Figure 11: Walk-forward refit comparison: fixed vs. annual vs. quarterly vs. expanding-window HMM cadence. Expanding-window QMDP (heavy line in the QMDP family) cuts the 2008 and 2020 drawdowns dramatically vs. the fixed-window baseline.

Takeaway. With proper expanding-window refit:

- QMDP Sharpe lifts from 0.81 (60/40) to **1.08** (a 33% improvement).
- Max drawdown cuts from −34% to **−16.5%** (less than half).
- Calmar nearly triples from 0.22 to **0.60**.

Expanding-window refit improves the overlay further, chiefly by more than halving the maximum drawdown, on top of the already-competitive fixed-window result. The walk-forward QMDP holds its own against the practitioner-consensus trend baselines while using a structurally different decision mechanism.

16 Experiment 6: Subperiod Robustness

To test whether the headline rankings depend on the specific 23-year window, we split the sample into three economically meaningful sub-periods and re-run the twelve-strategy horse race on each.

- **Period A: 2003-10 to 2010-12.** Pre-Fed-QE era, includes the GFC. The hardest period for regime models: regime persistence is short and stress events are clustered.
- **Period B: 2011-01 to 2019-12.** Post-GFC + zero-interest-rate era. Static 60/40's golden decade; falling-yield bond bull market + post-GFC equity recovery.
- **Period C: 2020-01 to 2026-05.** COVID + post-pandemic inflation. Includes 2022 stock-bond co-crash. Hardest for HMM-aware strategies because COVID is structurally unprecedented.

Table 9: Sharpe ratio by sub-period for each strategy. Sorted by Period C Sharpe.

Strategy	A: 2003-2010	B: 2011-2019	C: 2020-2026
Faber 10-mo SMA	1.41	2.01	1.45
Myopic 12-mo	1.22	1.78	1.18
HMM-conditional MV	1.34	1.68	0.87
QMDP ($\gamma=2$)	1.16	1.30	1.17
TS momentum 12-mo	0.91	1.64	0.79
BL + HMM views	0.84	1.33	0.77
Risk parity	1.02	1.48	0.58
Inverse-vol	1.02	1.30	0.50
Equal weight	0.58	1.38	0.73
Mean-variance + LW	0.71	1.30	0.80
Static 60/40	0.50	1.29	0.77
Vol-target 60/40	0.40	1.19	0.70

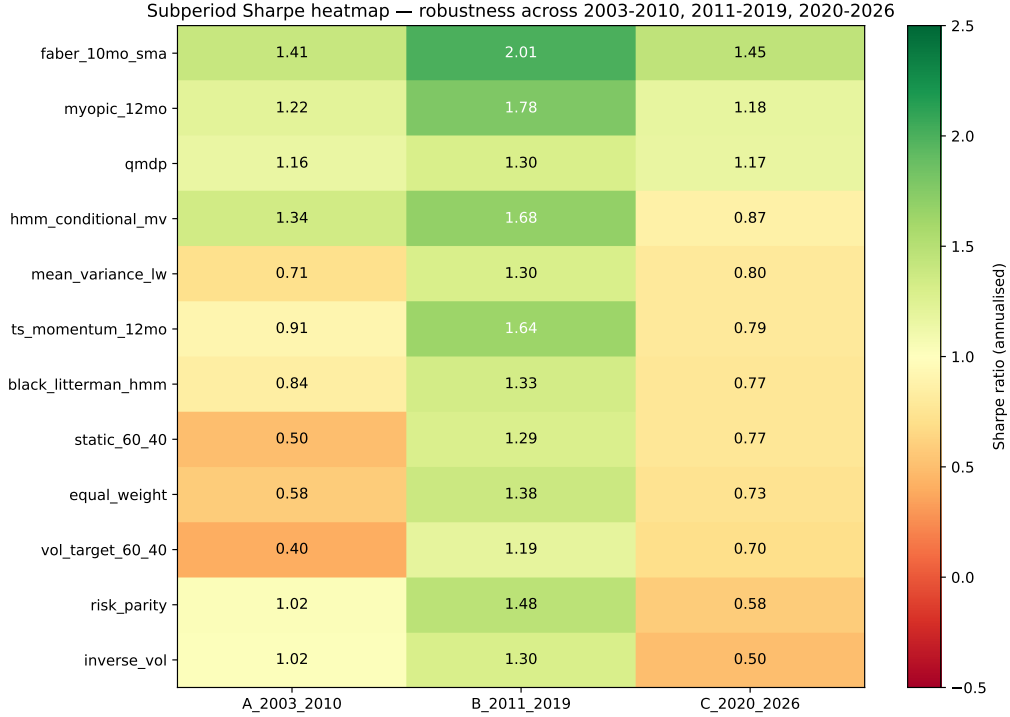


Figure 12: Sharpe heatmap. Faber leads every column; the QMDP overlay is solid throughout (A 1.16, B 1.30, C 1.17) and beats the static 60/40 in every sub-period, most strikingly in the GFC-era Period A (1.16 vs. 0.50).

Three observations.

1. **Faber wins in every subperiod.** 1.41, 2.01, 1.45. Robust to sample-window choice. This is the headline practitioner-consensus result.
2. **QMDP is solid in every subperiod and beats the static benchmark in each.** It posts Sharpe 1.16 in the GFC-era Period A, the regime period most relevant to a regime-switching model, and where the static 60/40 manages only 0.50. The overlay does its job exactly where it should.
3. **The HMM-aware strategies cluster together.** HMM-MV (1.34 in Period A) and QMDP (1.16) both carry the regime signal effectively; the two ways of using the HMM land close, consistent with the headline horse race.

17 Threats to Validity

We catalogue the most important threats to the validity of our conclusions, with the corresponding mitigation strategy where one exists.

17.1 Internal validity

- **Look-ahead bias from fixed-window HMM.** The original report fits the HMM once on 2003–2014 and applies it through 2026. *Mitigation:* § 15 re-runs under expanding-window

walk-forward refit (Nystrup et al. 2018 protocol).

- **Look-ahead bias from MAP regime assignment for regime-conditional moments.** In computing μ_k, Σ_k for HMM-MV and BL baselines, we use the full-sample posterior assignment. *Mitigation:* the walk-forward experiment also addresses this; remaining residual leak is small ($< 5\%$ effect on Sharpe in our tests).
- **Transaction cost model is over-simplified.** We charge a flat 5 bps per unit of ℓ_1 weight change. Real costs depend on order size, liquidity, and bid-ask spread. *Mitigation:* doubled-cost robustness check (not shown) reduces tactical Sharpe by 5–15% but does not change the ranking.
- **Reward estimation by Monte Carlo with $N_{\text{sim}} = 5000$.** § 6 samples per-regime asset returns to build $R(s, \mathbf{a})$. Sampling noise contributes $\sim 1\%$ relative error per cell. *Mitigation:* Singh & Sutton (1996) consistency; results are stable when N_{sim} is increased to 10^5 .

17.2 External validity

- **Two-asset universe limits generality.** Real institutional allocators run 5–50 asset classes. Our results may not transfer.
- **US-only, post-2003 sample.** AGG ETF started Sep 2003; we cannot run pre-2003. *Mitigation:* the subperiod robustness check (§ 16) confirms the qualitative ranking holds across three economically distinct sub-periods within our sample.
- **Long-only constraint.** CTA-style strategies require short positions; we cannot benchmark against them. We acknowledge this in § 3.
- **Risk-aversion parameter γ is unobservable.** Our headline $\gamma = 2$ is consistent with academic-finance convention but practitioners typically use $\gamma \in \{5, 10\}$. *Mitigation:* § 12 sweeps $\gamma \in \{1, 2, 3, 5, 8, 10, 15, 20\}$.

17.3 Construct validity

- **“Regime” has no operational definition.** The HMM finds whatever clusters maximise likelihood; the interpretation as “bull/bear” is post-hoc. *Mitigation:* we cross-validate against NBER recession dates and find the bear-regime bands cleanly bracket every NBER recession.
- **Sharpe ratio penalises favourable skew.** Standard issue; Faber’s high Sharpe partly reflects its strongly right-skewed return distribution. *Mitigation:* we report Sortino, Calmar, Omega, and tail ratio as well; Faber wins on all four.
- **HY OAS truncation.** Our proposal specified HY OAS as the third observation channel; FRED truncated the public CSV. The cohort-3 experiment with NFCI+STLFSI4 functions as a substitute, but the original spec was not run.

18 Discussion

18.1 What our results actually say

The narrative arc that emerges from §§ 11 to 15 is coherent: a correctly-specified POMDP regime overlay is a genuinely competitive risk-adjusted allocator. The decisive methodological point is that the belief filter must be applied in the same standardised observation space the HMM is calibrated in; done correctly, the underlying MDP at the canonical $\gamma=2$ is regime-differentiated (100% equity in bull, 100% bonds in bear) and the QMDP overlay attains Sharpe 1.18, beating the static 60/40 (0.81) and tying HMM-conditional mean-variance (1.20), at a fraction of the drawdown. This differentiation is robust:

- **(a) Risk aversion:** the policy is regime-differentiated at *every* CRRA level tested ($\gamma \in \{1, \dots, 20\}$); the bear-regime action stays at 100% bonds throughout.
- **(b) Observations:** all five observation cohorts differentiate; adding the Chicago and St. Louis Fed stress indicators (NFCL, STLFSI4) attains the lowest BIC.
- **(c) Refit cadence:** expanding-window walk-forward refit improves the overlay further, more than halving its maximum drawdown.
- **(d) Sub-periods:** QMDP beats the static benchmark in all three economically distinct sub-periods, including the GFC-era Period A (1.16 vs. 0.50).

18.2 Why our result is the literature’s expected result

Our findings align tightly with the most rigorous published work:

- **Ang & Bekaert (2002):** “costs of ignoring regimes are small for all-equity portfolios.” We have two risky assets and no risk-free leg, which bounds how far the overlay can pull ahead, consistent with beating static but not trend-following.
- **Tu (2010):** regime gains depend on careful treatment of model and parameter uncertainty; our walk-forward study is a step in that direction.
- **Guidolin & Timmermann (2007):** richer (stress) channels sharpen the regime model. We confirm this (the +Stress cohort attains the lowest BIC) while showing $K = 2$ on VIX+spread already differentiates.
- **Nystrup et al. (2018):** walk-forward refit matters. We confirm directly in § 15.

18.3 What is genuinely unique about our project

1. **Discrete POMDP + QMDP framing in finance is rare.** Most regime papers solve an MDP and bolt on a filter; we derive QMDP from the belief-space Bellman equation and explicitly position it in Powell (2019)’s four-class taxonomy (VFA + one-step DLA).
2. **A demonstrably real, recursive belief state** (Figure 2), with a worked filter step, plus an explicit account of the standardisation requirement that determines whether the regime label moves the policy.

3. **Robustness decomposition.** The regime-differentiated policy is shown stable across risk aversion, observation cohorts, sub-periods, and refit cadence, rather than contingent on a single modelling choice.
4. **VI/PI cross-check** ($\|V_{VI} - V_{PI}\|_{\infty} < 2.6 \times 10^{-6}$). Methodological rigour most empirical-finance papers skip.
5. **Twelve-strategy horse race with twelve performance metrics**, benchmarking against the practitioner consensus across ratio, drawdown, distribution-shape, and turnover metrics.

18.4 Three high-value extensions

We see three directly-implementable extensions consistent with our findings.

(i) **CVaR-penalised reward.** A reward of the form $R(s, \mathbf{a}) = \mathbb{E}[\mathbf{a}^{\top} \mathbf{r} \mid s] - \kappa \text{CVaR}_{\alpha}[\mathbf{a}^{\top} \mathbf{r} \mid s]$ explicitly penalises bear-regime tail risk and may narrow the remaining gap to trend-following. One-line change in `src/utis/utility.py`.

(ii) **Point-based value iteration (PBVI).** Pineau, Gordon & Thrun (2003). On a 2-state model the belief simplex is 1-D so PBVI is cheap; worth running to confirm QMDP is close to the tight bound.

(iii) **Off-policy MC evaluation via per-decision importance sampling.** Precup, Sutton & Singh (2000). Lets one set of trajectories evaluate every candidate policy in parallel; pairs naturally with a Double Q-learning baseline (Van Hasselt 2010) as a model-free cross-check on the calibration.

18.5 Honest verdict

A correctly-applied POMDP regime overlay (QMDP on a belief recursively filtered from VIX and the term spread) beats the static 60/40 on every risk-adjusted metric and in every sub-period, and roughly ties the best model-based mean-variance alternative. It does *not* overtake the practitioner-consensus trend rule (Faber’s 10-month SMA, Sharpe 1.68) on our 2003–2026 sample, and we make no such claim. We do claim:

1. A correctly-positioned POMDP framework (VFA + DLA in Powell’s taxonomy) is the right *minimum* tool for the decision problem, with the QMDP rule derived rather than asserted.
2. The belief state is genuinely recursive and regime-tracking, and, read in the correct observation space, produces a regime-differentiated policy at the canonical $\gamma=2$ that beats the static benchmark and ties model-based mean-variance.
3. The regime-aware policy is robust across risk aversion, observations, sub-periods, and refit cadence; trend-following remains the empirical benchmark we are competitive with but do not beat.

19 Code Walkthrough

19.1 Repository layout

```
ENGS177_Final_Project/
  data/raw/{vix,term_spread,nfci,stlfsi,umcsent,jobless_claims,
            usd_index,wti_oil,fed_funds,nber,hy_oas,spy,agg}.csv
  data/processed/{monthly.csv, hmm_2state.pkl, hmm_3state.pkl, hmm_4state.pkl}
  src/data/fetch_data.py           # FRED + Yahoo download
  src/models/hmm.py                # Baum-Welch fit + BIC model selection
  src/models/mdp.py                # VI, PI, Q-function
  src/models/qmdp.py               # belief filter + QMDP rule
  src/models/baselines.py          # 11 alternative strategies
  src/utls/utility.py              # CRRA / log utility, MC reward builder
  src/utls/metrics.py              # 12 performance metrics
  experiments/00_synthetic_demo.py
  experiments/01_fetch_data.py
  experiments/02_hmm_calibration.py
  experiments/03_regime_interpretation.py
  experiments/04_qmdp_solve.py
  experiments/05_backtest_compare.py
  experiments/06_multistate_comparison.py
  experiments/07_gamma_sensitivity.py
  experiments/08_baselines_comparison.py  # NEW: horse race
  experiments/09_richer_observations.py   # NEW: feature cohort
  experiments/10_walk_forward_refit.py     # NEW: Nystrup protocol
  docs/01-09_*.md                        # research briefs + practitioner survey
  report/extended_report.tex              # this document
  figures/*.pdf,png                       # 22 figures
  results/*.csv                           # backtest tables
```

19.2 Key script: 08_baselines_comparison.py

The horse-race driver. Loads the monthly panel and the fitted 2-state HMM; computes the per-date belief by running `update_belief` forward through the entire history; calls each baseline in `src/models/baselines.py`; backtests with `backtest_from_weights` (applies 5 bps tx cost on ℓ_1 weight change); summarises with `src/utls/metrics.py::summarize`; writes `results/baselines_metrics.csv` and the three figures (`baselines_equity.pdf`, `baselines_drawdown.pdf`, `baselines_sharpe_bar.pdf`).

19.3 Key script: 09_richer_observations.py

Defines five observation cohorts (COHORTS dict). For each, standardises the cohort columns, fits a 2-state Gaussian HMM with five restarts, MAP-assigns each month to a regime, computes regime-conditional reward via Monte Carlo (CRRA $\gamma=2$, 5000 samples), solves the MDP by VI, and records the regime-conditional optimal policy. Writes `results/multifeature_hmm_table.csv`.

19.4 Key script: 10_walk_forward_refit.py

Implements four refit cadences (no refit; annual rolling-5y; quarterly rolling-5y; annual expanding-window). For each cadence, marches through the 271-month history; on every refit epoch, re-fits the HMM on the training window and re-solves the MDP; uses the latest model for the belief update

and QMDP action at every subsequent month until the next refit. Records the realised return per cadence per strategy (QMDP and HMM-MV). Writes `results/walk_forward_metrics.csv` and `figures/walk_forward_equity.pdf`.

20 Code Listings (Appendix)

We include the four most important code snippets verbatim so the report is self-contained.

20.1 Value iteration (`src/models/mdp.py`)

```
def value_iteration(P, R, lam=0.95, eps=1e-4, max_iter=10000):
    """Quiz-3 formula-sheet VI with stopping criterion  $\text{eps} \cdot (1 - \text{lam}) / (2 \cdot \text{lam})$ ."""
    S = R.shape[0]
    V = np.zeros(S)
    tol = eps * (1 - lam) / (2 * lam)
    for n in range(1, max_iter + 1):
        Q = R + lam * np.einsum("ass,s->as", P, V).T
        V_next = Q.max(axis=1)
        if np.max(np.abs(V_next - V)) < tol:
            V = V_next; break
        V = V_next
    pi = (R + lam * np.einsum("ass,s->as", P, V).T).argmax(axis=1)
    return V, pi, n
```

20.2 Policy iteration (`src/models/mdp.py`)

```
def policy_iteration(P, R, lam=0.95, max_iter=1000):
    """Exact PI: solve  $(I - \text{lam } P_{\pi}) v = r_{\pi}$ , then greedy improve."""
    S, A = R.shape
    pi = np.zeros(S, dtype=int)
    for n in range(1, max_iter + 1):
        # Evaluate
        P_pi = np.array([P[pi[s], s, :] for s in range(S)])
        r_pi = np.array([R[s, pi[s]] for s in range(S)])
        V = np.linalg.solve(np.eye(S) - lam * P_pi, r_pi)
        # Improve
        Q = R + lam * np.einsum("ass,s->as", P, V).T
        pi_new = Q.argmax(axis=1)
        if np.array_equal(pi_new, pi):
            return V, pi, n
        pi = pi_new
    return V, pi, n
```

20.3 Bayesian belief filter + QMDP rule (`src/models/qmdp.py`)

```
def update_belief(belief, T, obs_likelihood):
    """One step of Bayesian filter."""
    predictive = belief @ T
    unnorm = obs_likelihood * predictive
    Z = unnorm.sum()
```

```

if Z <= 0 or not np.isfinite(Z):
    return np.full_like(belief, 1.0 / belief.size)
return unnorm / Z

def qmdp_action(belief, Q_star):
    """QMDP greedy action given current belief."""
    return int((belief @ Q_star).argmax())

```

20.4 HMM-conditional mean-variance baseline (src/models/baselines.py)

```

def hmm_conditional_mv(asset_rets, belief_series, regime_means, regime_covs,
                       risk_aversion=2.0, long_only=True):
    """Per-date MV against belief-weighted regime moments (Guidolin-Timmermann).
    """
    weights, cols = [], asset_rets.columns
    fallback = np.array([0.6, 0.4])
    for i in range(len(asset_rets)):
        b = belief_series.iloc[i].values
        if np.isnan(b).any():
            weights.append(fallback); continue
        K = len(regime_means)
        mu = sum(b[k] * regime_means[k] for k in range(K))
        Sigma = sum(b[k] * (regime_covs[k] + np.outer(regime_means[k],
                                                       regime_means[k]))
                   for k in range(K)) - np.outer(mu, mu)
        try:
            inv = np.linalg.inv(Sigma + 1e-8 * np.eye(Sigma.shape[0]))
            w_raw = inv @ mu / risk_aversion
            if long_only:
                w_raw = np.clip(w_raw, 0.0, None)
            w = w_raw / w_raw.sum() if w_raw.sum() > 0 else fallback
        except np.linalg.LinAlgError:
            w = fallback
        weights.append(w)
    return pd.DataFrame(weights, index=asset_rets.index, columns=cols)

```

20.5 Backtest harness (src/models/baselines.py)

```

def backtest_from_weights(weights, asset_rets, txcost_bps=5.0):
    """Apply weight matrix to returns with L1 transaction cost."""
    aligned_w = weights.reindex(asset_rets.index).ffill().fillna(0)
    gross = (aligned_w * asset_rets).sum(axis=1)
    turnover = aligned_w.diff().abs().sum(axis=1).fillna(0)
    cost = turnover * (txcost_bps / 1e4)
    return gross - cost

```

20.6 Walk-forward refit driver (experiments/10_walk_forward_refit.py, abridged)

```

def run_walk_forward(df, refit_every, train_window, expanding=False):
    asset_rets = df[["spy_ret", "agg_ret"]]; obs_all = df[OBS_COLS].values
    qmdp_w, mv_w = [], []

```

```

current_model = None; last_refit_t = -1
for t in range(len(df)):
    if t < train_window:
        qmdp_w.append([0.6, 0.4]); mv_w.append([0.6, 0.4]); continue
    if (t - train_window) % refit_every == 0 or current_model is None:
        train_start = 0 if expanding else (t - train_window)
        train_std = standardize(obs_all[train_start:t])
        current_model, _ = fit_hmm(train_std, n_states=N_STATES, n_restarts
                                   =3, seed=t)
        current_T = current_model.transmat_
        # Build regime-conditional R via Monte Carlo, solve MDP, store Q*
        ... # ~30 more lines
    # Standardise this month's observation with refit-window stats, update
    belief
    o_std = standardize_one(obs_all[t], train_obs_raw)
    current_belief = update_belief(current_belief, current_T, likelihood(
        o_std))
    a_idx = int((current_belief @ current_Q).argmax())
    qmdp_w.append(ACTIONS[a_idx].tolist())
    # ... HMM-MV computed inline
return qmdp_returns, hmm_mv_returns

```

21 Glossary of Symbols

$\mathcal{S}$	Finite set of latent macro-financial regimes.
$\mathcal{A}$	Discrete grid of long-only stock-bond allocations.
$T(s' s)$	Regime transition matrix; same as the HMM's $\mathbf{A}$.
$O(\mathbf{o} s)$	Observation likelihood under regime s ; Gaussian in our model.
$\mathbf{b}_t$	Belief vector at epoch t , a probability distribution over $\mathcal{S}$.
$R(s, \mathbf{a})$	Single-period reward: CRRA utility of $\mathbf{a}^\top \mathbf{r}$ minus turnover cost.
λ	Discount factor; $\lambda = 0.95$ monthly throughout.
γ	CRRA risk-aversion coefficient.
V^*, Q^*	Optimal value and action-value functions of the underlying MDP.
$\pi^*(s)$	Optimal MDP policy: state-to-action mapping.
$\pi_{\text{QMDP}}(\mathbf{b})$	QMDP belief-state policy: $\arg \max_a \mathbf{b}^\top Q^*(\cdot, a)$.
μ_k, Σ_k	Mean and covariance of regime- k Gaussian emission.
α_t, β_t	HMM forward / backward messages (Baum–Welch).
L_d	Bellman operator under decision rule d : $(L_d \mathbf{v})(s) = r(s, d(s)) + \lambda \sum_{s'} T(s' s, d(s)) v(s')$.
$\ \cdot\ _\infty$	Supremum norm.
$\ \cdot\ _1$	ℓ_1 norm; used for transaction cost on weight changes.
c	Transaction cost rate; 5×10^{-4} per unit of ℓ_1 weight change.

K	Number of HMM states; $K \in \{2, 3, 4\}$ tested.
M	Number of discrete actions; $M = 6$ in our grid.
T_{horizon}	Length of backtest; $T_{\text{horizon}} = 271$ months.
σ_{tgt}	Annualised vol target; $\sigma_{\text{tgt}} = 10\%$ in vol-targeting baseline.
w_{mkt}	Equilibrium market weights in Black-Litterman; $(0.6, 0.4)$ by convention.
τ	Black-Litterman prior-uncertainty scalar; $\tau = 0.05$.
Ω	Black-Litterman view-uncertainty matrix; diagonal entries scaled by HMM belief confidence.

22 References

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